

Which Optimizers Find the Support?

An Empirical Comparison on Sparse-Support Tasks

Mahlia Merville-Hipeau, François Goybet, Matteo Pinto – GitHub: `optiML_support`
Optimization for Machine Learning – EPFL CS-439, Spring 2026

Abstract—Recent work has shown that mini-batch SGD can drive neural networks to identify the low-dimensional *support* of a target function in their first layer through an implicit regularization effect. However, it remains an open question whether this feature-selection mechanism persists under modern optimization algorithms. In this report, we compare the support-identification capabilities of full-batch Gradient Descent (GD), mini-batch SGD, Adam, and Muon on sparse-support regression tasks. We evaluate these optimizers across linear, sine, and staircase target functions where relevant and irrelevant features are decorrelated in expectation. Our results show that support recovery is highly optimizer-dependent. While small-batch SGD successfully isolates relevant coordinates and suppresses irrelevant ones, Adam achieves the lowest predictive loss but fails to cleanly identify the support. Finally, we observe that Muon achieves the highest loss and does not identify the support.

I. INTRODUCTION

A trained neural network must identify the *support* of its target function, the subset of input coordinates that actually influence the output, to learn an interpretable and robust representation. Beneventano et al. [1] show that this identification typically happens in the *first layer*, but only under specific optimization regimes: full-batch gradient descent (GD) approximates the target well, yet its first-layer weights interacting with irrelevant input coordinates do not converge to zero. Mini-batch SGD, in contrast, drives those weights to zero. The paper attributes this to a second-order implicit regularization term that arises from the discrepancy between the mini-batch and full-batch input covariance matrices.

The original study compares GD, SGD at three batch sizes, and GD with weight decay. Several optimizers that are standard in modern deep learning such as Adam and the recent Muon optimizer [2], are not studied. They differ from SGD in important ways: Adam [3] uses coordinate-wise adaptive learning rates and Muon orthogonalizes momentum updates via Newton-Schulz iterations. Whether the support-finding mechanism described by [1] survives, disappears, or is replaced by a different mechanism under these optimizers is, to the best of our knowledge, an open empirical question.

Our contributions are threefold: (i) we reproduce the GD vs. SGD vs. GD+wd comparison of [1] on a sparse-support regression problem. (ii) we extend the comparison to Adam and Muon on three target functions (linear, sine, staircase). (iii) we summarize each optimizer with different metrics: loss, norm of W_1 , Gram matrix, irrelevant norm ratio and singular values.

II. RELATED WORK

Mini-batch SGD selects different minima from full-batch GD, typically favoring flatter regions of the loss landscape that generalize better [4]. Beneventano et al. [1] recently showed that on finite datasets, first-layer support identification is an artifact of the optimizer rather than of the target function. This complements a separate line of work on support recovery in infinite-data regimes [5]. Whether modern optimizers preserve this property is largely open. Adam has been argued to behave closer to sign descent than to noisy SGD, with corresponding differences in implicit bias [6], and the recent Muon optimizer replaces the gradient step on matrix parameters with an approximately orthogonal update [2]. To our knowledge, no prior work compares these optimizers on our studied task.

III. EXPERIMENTAL SETUP

A. Sparse-support synthetic data

Following [1], we generate a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{2n}$ with $n = 5000$, input dimension $d = 15$, and $r = 5$ *relevant* coordinates. Inputs $x \in \mathbb{R}^d$ are drawn from $\mathcal{N}(0, I_d)$. We follow Assumption 1 of [1]: the expectation over the dataset of relevant times irrelevant features is 0, making them decorrelated in expectation.

The target depends only on the relevant coordinates $x[:r]$:

$$y(x) = \begin{cases} \sum_{i=1}^r x_i & \text{(linear),} \\ \sin(\sum_{i=1}^r x_i) & \text{(sine),} \\ \sum_{i=1}^r x_i^i & \text{(staircase),} \end{cases} \quad (1)$$

To match the protocol of [1], each x_i is duplicated with labels $y_i \pm \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ and $\sigma = 10^{-3}$ to force residuals to oscillate around zero even at a perfect linear fit, which is what powers the second-order SGD effect of [1].

B. Model

We use a fully connected MLP with three hidden layers of width 15 and a scalar output, without biases. This matches the linear-network setting in which Theorem 1 of [1] is stated. All hyperparameters are reported in Table IV.

C. Optimizers compared

We compare six optimizer configurations. The first four (GD, SGD at two batch sizes, GD+wd of 0.1) reproduce the comparison of [1], with GD+wd serving as an explicitly regularized reference baseline. The last two (Adam, Muon) extend it. For each optimizer, we compare the final first-layer structure to its initialization.

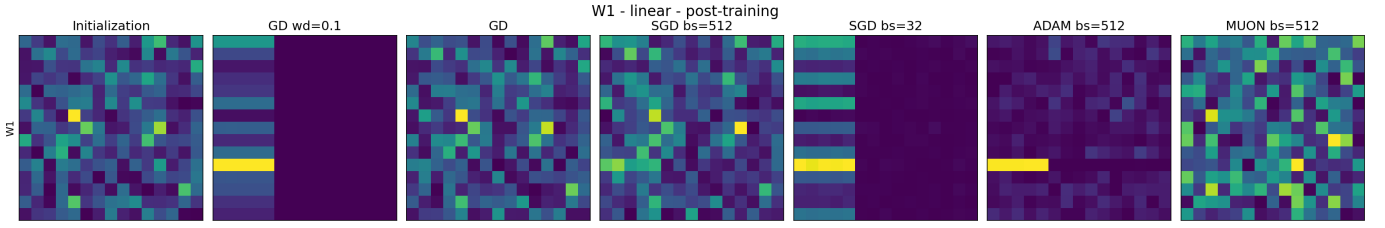


Fig. 1. First-layer weights W_1 (absolute values) and Gram matrices $W_1^T W_1$ (min-max normalized) at convergence, linear target (Eq. 1), $r = 5$, $d = 15$. SGD-32 and GD+wd zero out the irrelevant block (dark right half); Adam achieves the lowest loss solution but does not suppress irrelevant weights; Muon fails to converge.

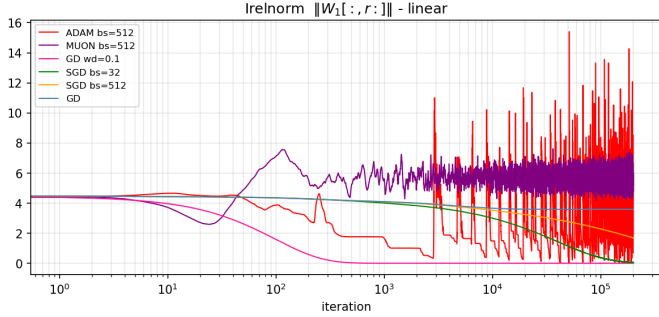


Fig. 2. Norm of irrelevant first-layer weights $\|W_1[:, r:d]\|_F$ over training iterations, on the linear target (Eq. 1), $r=5$, $d=15$.

D. Metrics

During training we record the MSE loss to verify that each optimizer reaches comparable fit. We simultaneously track the Frobenius norm of the irrelevant block $\|W_1[:, r:d]\|_F$, whose trajectory distinguishes optimizers that shrink irrelevant weights monotonically (like SGD) from those that stall at a positive value (like GD). At convergence we add three diagnostics on the first layer. The normalized ratio $\|W_1[:, r:d]\|_F / \|W_1\|_F$ controls for overall weight scale, ensuring uniform shrinkage is not mistaken for genuine support selection. The Gram matrix $W_1^T W_1$ visualizes which input coordinates the first layer attends to: a clean $r \times r$ top-left block confirms that the support has been isolated. Finally, the singular spectrum of W_1 shows whether mass concentrates in a few directions.

IV. RESULTS

A. GD and SGD

We first consider the linear target. We report the final loss in Table I, the irrelevant norm dynamics in Figure 2, the irrelevant norm ratio at convergence in Table II, and the first-layer weights with Gram matrices in Figure 1. Our results reproduce the main finding of [1]. Full-batch GD reaches a training loss of 0.033 but fails to identify the support in the first layer: the irrelevant norm decreases only marginally from 4.47 to 3.59 (ratio: 78.5%), the Gram matrix shows no block structure, and the singular value spectrum remains spread across all directions, confirming that GD does not concentrate the support in W_1 .

Mini-batch SGD shows the batch-size dependence predicted by Theorem 1 of [1]. With $b=512$, the irrelevant norm drops from 4.47 to 1.69 (ratio: 69.2%). With $b=32$, it collapses to 0.05 (ratio: 3.5%), the Gram matrix displays a clear $r \times r$ block structure, and the singular value spectrum concentrates on a single dominant direction (Figure 5), confirming that the first layer has identified the support. GD with weight decay ($\lambda=0.1$) achieves the strongest suppression, driving the irrelevant norm from 4.43 to 0.001 (ratio: 0.1%) with a final loss of 0.001 (Table II), consistent with Proposition 3 of [1].

The same qualitative pattern holds on the sine and staircase targets, as shown in Figures 6–11 in the Appendix: SGD with $b=32$ consistently achieves the strongest irrelevant-norm suppression among the non-regularized optimizers, while GD fails to identify the support in W_1 across all three targets.

B. Adam

Adam achieves the lowest final loss on the linear target (0.0004), outperforming all other configurations including GD+wd (0.001), as reported in Table I. However, its irrelevant norm decreases from 4.44 to 1.91 (ratio: 38.3%, Table II), intermediate between GD (78.5%) and SGD-32 (3.5%): Adam partially suppresses irrelevant weights but does not cleanly identify the support in W_1 (Figure 1). The singular value spectrum (Figure 5) reflects this ambiguity: the first singular value is substantially amplified relative to initialization, but the remaining directions follow a decay pattern similar to initialization rather than collapsing to zero, indicating that W_1 has not converged to a low-rank solution. On the sine target the suppression weakens (ratio: 63.0%), and on the staircase target the irrelevant norm increases to 6.39 (ratio: 50.8%) (Table II).

C. Muon

Muon reaches a final loss of 0.408 on the linear target, far above GD (0.033) and all SGD variants (Table I). The irrelevant norm increases from 4.46 to 5.97 (ratio: 81.9%, Table II), the Gram matrix shows no block structure (Figure 1), and the singular value spectrum remains spread across all directions with several large values (Figure 5), confirming that W_1 does not concentrate on the relevant features.

On the sine and staircase targets (Figures 6–11 in the Appendix), the pattern worsens: the irrelevant norm reaches 4.57 (ratio: 80.8%) on sine and 7.47 (ratio: 65.0%) on staircase, with losses of 0.575 and 231.3 respectively (Table II). Across

all three targets, Muon consistently increases the irrelevant norm above its initialization value, and the Gram matrix displays no block structure.

TABLE I
FINAL TRAINING LOSS FOR TARGET FUNCTIONS ACROSS OPTIMIZERS.

Target	GD	SGD ($b=512$)	SGD ($b=32$)	Adam	Muon	GD+wd
Linear	0.033	0.0390	0.0228	0.0004	0.4082	0.0009
Sine	0.4578	0.4805	0.3580	0.5063	0.5751	0.4944
Staircase	657.3192	226.3478	84.0119	291.2697	231.2776	655.9221

TABLE II
IRRELEVANT NORM RATIO $\|W_1[:, r :]\|_F / \|W_1\|_F$ (%) AT CONVERGENCE ACROSS OPTIMIZERS AND TARGETS.

Target	GD	SGD ($b=512$)	SGD ($b=32$)	Adam	Muon	GD+wd
Linear	78.5%	69.2%	3.5%	38.3%	81.9%	0.1%
Sine	78.7%	74.0%	28.8%	63.0%	80.8%	0.0%
Staircase	78.9%	8.3%	6.3%	50.8%	65.0%	8.7%

V. DISCUSSION

A. Support identification across optimizers

Our experiments confirm that support identification is an optimizer-dependent phenomenon. On the linear target, full-batch GD decreases the training loss but keeps most of its first-layer mass on irrelevant coordinates. In contrast, SGD with a small batch size clearly suppresses the irrelevant block of W_1 ; for $b = 32$, the irrelevant norm ratio drops to 3.5% (Table II), and the Gram matrix shows a clear block structure on the first r coordinates (Figure 3). This matches the mechanism described by Beneventano et al. [1]: mini-batch stochasticity induces an implicit regularization effect on the first layer. This explains why SGD-32 identifies the support much more cleanly than SGD-512.

B. Adam and support identification

Adam gives a different tradeoff than SGD. On the linear target, it achieves the lowest final loss, showing that it finds a highly predictive solution. However, this solution does not identify the support cleanly: its irrelevant norm ratio remains much larger than SGD-32 or GD+wd (Table II), and the Gram matrix does not show the same sharply defined relevant block (Figure 3). This suggests that Adam’s implicit bias is better aligned with fast loss minimization than with sparse first-layer feature selection.

Across the nonlinear targets, Adam remains less consistent than SGD-32. It does not obtain the best loss on the sine target, and on the staircase target it keeps a large irrelevant norm ratio despite improving over GD in loss (Figures 7, 10). Overall, Adam learns useful predictive directions, but does not reliably separate relevant from irrelevant input coordinates.

C. Muon as a failure case

Across the three target functions, Muon does not produce any clean relevant block in the Gram matrix (Figure 3), and the irrelevant norm remains large or even increases (Table II,

Figures 4, 7, 10). On the linear target, it also obtains a much worse final loss than all other optimizers.

This behavior is consistent with Muon’s update geometry. Muon orthogonalizes matrix updates, which encourages balanced movement across matrix directions. Support identification requires the opposite bias: the optimizer must concentrate the first layer on a small subset of input coordinates while suppressing the rest. Therefore, Muon’s inductive bias appears poorly aligned with sparse-support learning, which is what our experiments showcase. This should not be interpreted as a general failure of Muon, but rather as a mismatch between its update rule and this specific feature-selection task.

D. Limitations

Our experiments are restricted to synthetic data where relevant and irrelevant coordinates are decorrelated by construction, which makes support recovery easy to measure but does not reflect realistic data, where features are correlated and the notion of an irrelevant coordinate is less clear. Our diagnostics are also centered on the first layer, following Beneventano et al. [1]: this is informative for support identification, but for nonlinear targets a coordinate may remain active in W_1 while being used only weakly downstream. Extending the analysis to realistic datasets therefore remains open; our preliminary attempts on CIFAR-10 with pretrained backbones were inconclusive.

VI. CONCLUSION

In this work, we investigated how the choice of optimizer dictates a neural network’s ability to identify the relevant support of a target function in its first layer. By extending prior analyses to include Adam and Muon optimizers, we demonstrated that the implicit biases of these algorithms alter feature-selection dynamics.

Our experiments show that small-batch SGD reliably suppresses the irrelevant block of W_1 across the linear, sine, and staircase targets, consistent with the implicit regularization mechanism described by Beneventano et al. [1], and approaches the explicitly regularized GD+wd reference.

Adam reaches the lowest loss on the linear target but only partially shrinks irrelevant coordinates, and its inductive bias appears better aligned with fast fitting than with sparse feature selection. Muon fails on both axes: its orthogonalized matrix updates spread mass across input coordinates rather than concentrating it on the support, which we read as a mismatch between its update geometry and the sparse-support task rather than a general failure of the optimizer.

Together, these results show that low loss and clean support recovery are distinct outcomes, and the choice of optimizer determines which one is achieved. Whether this picture transfers to realistic settings with correlated features and deeper feature use remains an open empirical question.

REFERENCES

- [1] P. Beneventano, A. Pinto, and T. Poggio, “How neural networks learn the support is an implicit regularization effect of SGD,” *arXiv preprint arXiv:2406.11110*, 2024.
- [2] K. Jordan, Y. Jin, V. Boza, J. You, F. Cesista, L. Newhouse, and J. Bernstein, “Muon: An optimizer for hidden layers in neural networks,” 2024, <https://kellerjordan.github.io/posts/muon/>.
- [3] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” *arXiv preprint arXiv:1412.6980*, 2014.
- [4] N. S. Keskar, D. Mudigere, J. Nocedal, M. Smelyanskiy, and P. T. P. Tang, “On large-batch training for deep learning: Generalization gap and sharp minima,” in *International Conference on Learning Representations (ICLR)*, 2017.
- [5] A. Damian, J. D. Lee, and M. Soltanolkotabi, “Neural networks can learn representations with gradient descent,” in *Conference on Learning Theory (COLT)*, 2022.
- [6] F. Kunstner, J. Chen, J. W. Lavington, and M. Schmidt, “Noise is not the main factor behind the gap between SGD and Adam on transformers, but sign descent might be,” *International Conference on Learning Representations (ICLR)*, 2023.

APPENDIX

A. Linear Target

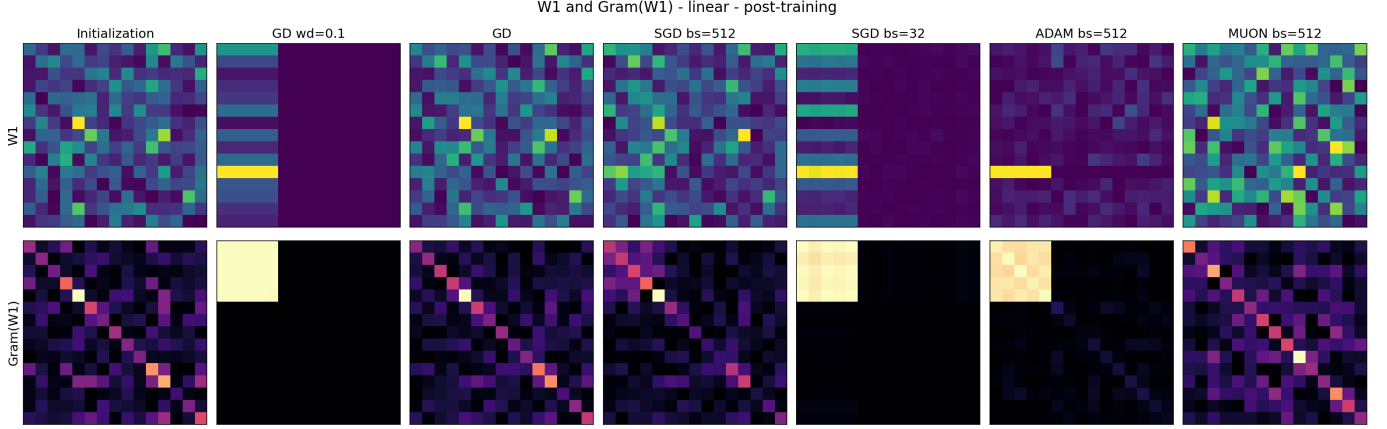


Fig. 3. First-layer weights W_1 (absolute values) and Gram matrices $W_1^\top W_1$ (min-max normalized) at convergence, linear target (Eq. 1), $r = 5$, $d = 15$.

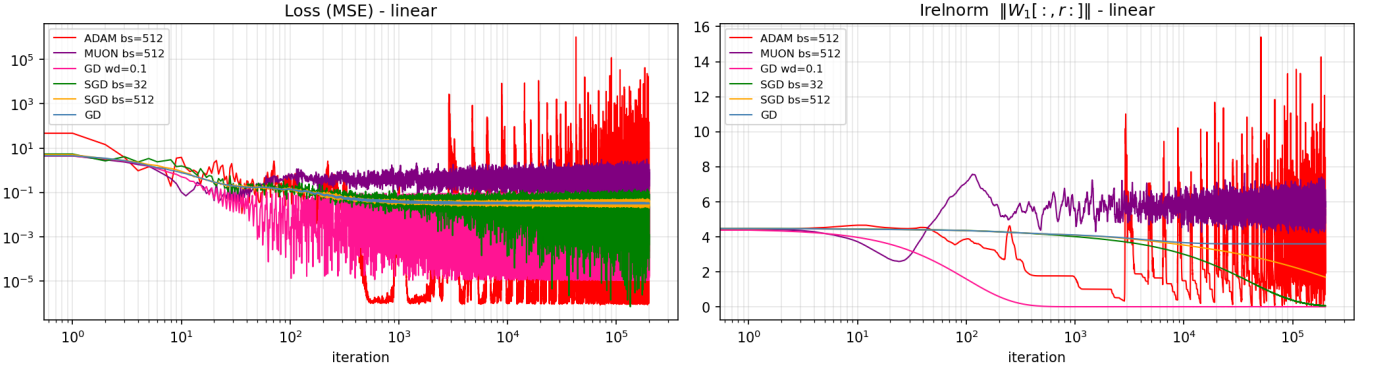


Fig. 4. Training loss (MSE, left) and norm of irrelevant first-layer weights $\|W_1[:, r:d]\|_F$ over training iterations (right), on the linear target (Eq. 1), $r=5$, $d=15$.

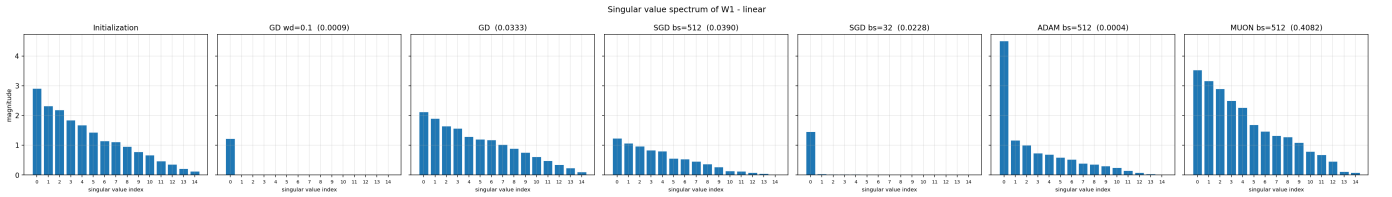


Fig. 5. Singular value spectrum of W_1 at convergence, linear target (Eq. 1), $r=5$, $d=15$.

B. Sine Target

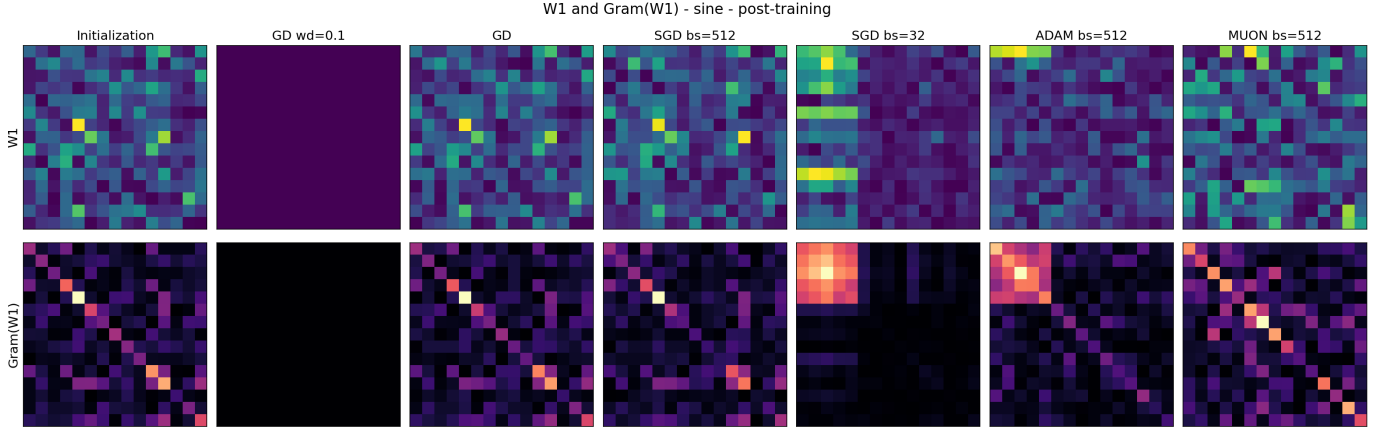


Fig. 6. First-layer weights W_1 (absolute values) and Gram matrices $W_1^T W_1$ (min-max normalized) at convergence, sine target (Eq. 1), $r=5$, $d=15$.

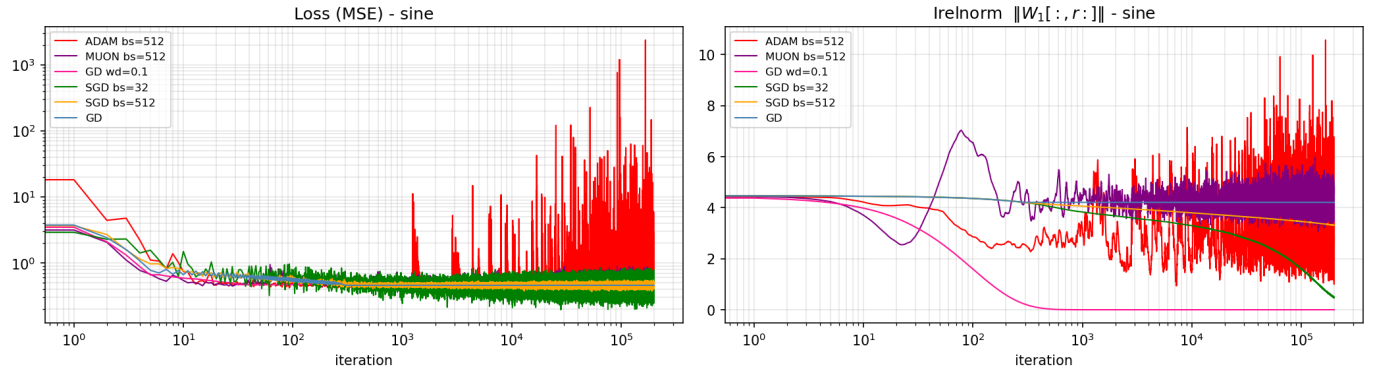


Fig. 7. Training loss (MSE, left) and norm of irrelevant first-layer weights $\|W_1[:, r:d]\|_F$ over training iterations (right), on the sine target (Eq. 1), $r=5$, $d=15$.

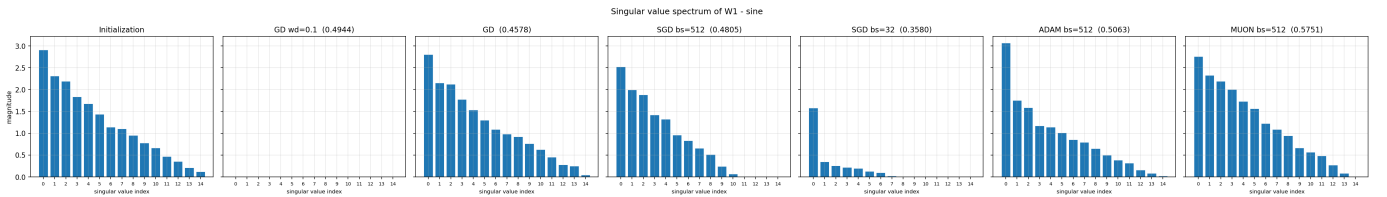


Fig. 8. Singular value spectrum of W_1 at convergence, sine target (Eq. 1), $r=5$, $d=15$.

C. Staircase Target

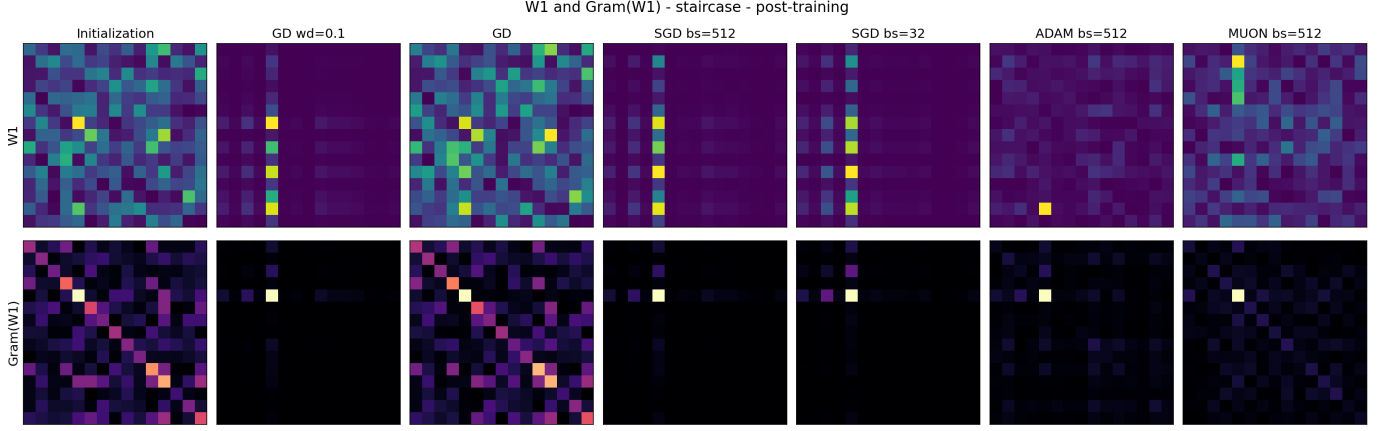


Fig. 9. First-layer weights W_1 (absolute values) and Gram matrices $W_1^T W_1$ (min-max normalized) at convergence, staircase target (Eq. 1), $r = 5$, $d = 15$.

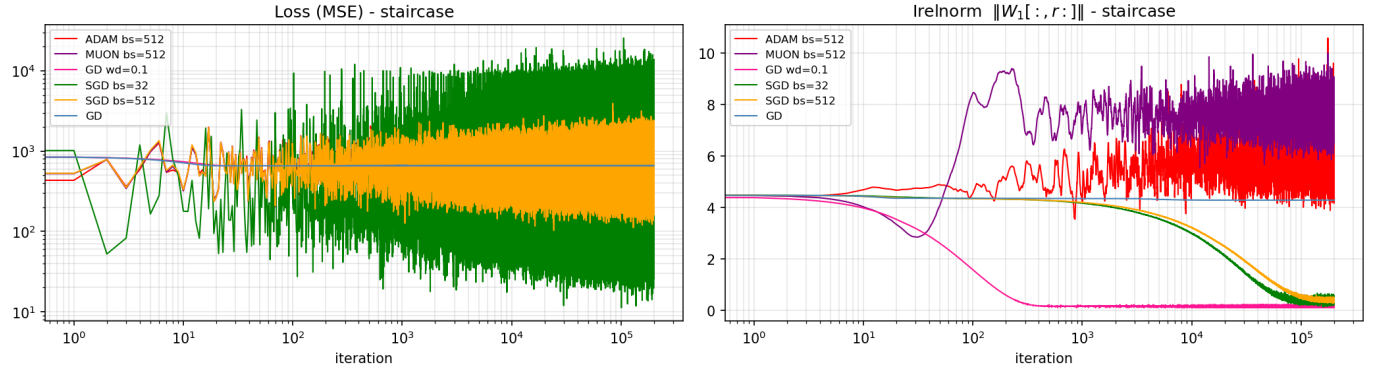


Fig. 10. Training loss (MSE, left) and norm of irrelevant first-layer weights $\|W_1[:, r:d]\|_F$ over training iterations (right), on the staircase target (Eq. 1), $r=5$, $d=15$.

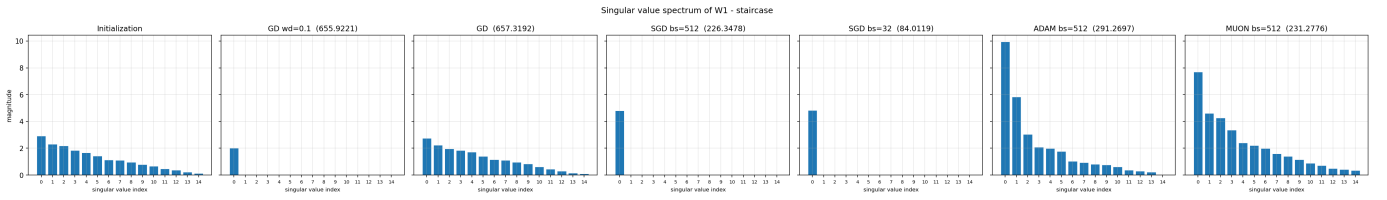


Fig. 11. Singular value spectrum of W_1 at convergence, staircase target (Eq. 1), $r=5$, $d=15$.

D. Additional tables

TABLE III
TRAINING TIME OF THE DIFFERENT MODELS.

Target	GD	SGD ($b=512$)	SGD ($b=32$)	Adam	Muon	GD+wd
Linear	3h01	56min	20min	11min	13min	3h03
Sine	3h	54min	20min	11min	13min	3h04
Staircase	3h02	1h03	23min	11min	12min	3h09

TABLE IV
HYPERPARAMETERS USED IN THE EXPERIMENTS. UNLESS STATED OTHERWISE, VALUES ARE SHARED ACROSS ALL RUNS.

Component	Hyperparameter value
Dataset size	$n = 5000$ base samples, doubled internally to 10000 examples
Input dimension	$d = 15$
Support size	$r = 5$ relevant coordinates
Noise level	$\sigma = 0.001$
Relevant coordinates	First r input coordinates
Irrelevant coordinates	Last $d - r$ centered coordinate-wise before training
Model	Bias-free MLP
Hidden widths	(15, 15, 15)
Activation	None
Weight initialization	He normal
Loss	Mean squared error
Gradient clipping	Global norm clipping with threshold 1.0
Default learning rate	0.1
Weight decay	0.1 for WD runs, 0 otherwise
GD batch size	Full doubled dataset, i.e. 10000 examples
SGD batch size	512, 32 for respective runs
Adam and Muon mini-batch size	512
Optimizer steps	200000
Muon momentum	0.95
Muon Newton–Schulz steps	5